MediScan Audit — Technical Report

Arogya Diagnostics: paediatric chest X-ray pneumonia screening

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Every table is copied from a file in `results/` written by a script in the repository (`python run_all.py`); every figure is a file in `figures/`. Seeds 0, 1, 2 are logged in `results/seed_log.txt`. Environment: Python 3.12, PyTorch 2.14.1 (CUDA 12.6), RTX 4050 laptop GPU.

1 Summary

- **The wrong instruction.** Instruction 2, "99% accuracy", is the wrong target. At the camp's 9:1 NORMAL:PNEUMONIA mix a tool that flags nobody is 90% accurate, and a 99%-accurate tool can still miss 10 of 100 sick children (cost 200 per 1 000 children) or raise 10 false alarms (cost 10): accuracy cannot tell these apart (Section 5, Table 10). Instructions 1 (evidence) and 3 (200 ms, 25 MB) were reasonable and are met (Section 11).
- **What the audit found.** A 16-image validation set (an observed 16/16 is compatible with a true accuracy as low as 79.4%); a class mix that shifts between splits (74% pneumonia in train, 62.5% in test, 50% in val); strong non-lung shortcuts (image size and aspect ratio); and at most modest near-duplicate leakage (15 of 624 test images flagged by a perceptual hash, and the five closest pairs are look-alikes, not copies). No patient overlap between train and test was found by file-name ID. The 98.4% cannot be reproduced or verified.
- **What we built.** ArogyaNet, a 242 319-parameter CNN trained from scratch. On a patient-disjoint locked test set it reaches ROC-AUC 0.989, sensitivity 94.9% and specificity 95.6% at the cost-optimal threshold: about 5 missed and 40 falsely flagged children per 1 000 at 10% prevalence.
- **How far it generalises.** On an external hospital (adult-dominated, PA view) ROC-AUC fell from 0.989 to 0.899 and specificity from 95.6% to 72.6% at the same threshold. The CLAHE fix did not help (AUC 0.852).
- **Honest limitations.** Pretrained MobileNetV3 and ResNet18 beat ArogyaNet in-distribution; the squeeze-excitation hypothesis was not supported; the 95% sensitivity target is not demonstrably met (95% CI about 93–97%).

Notation

Let TP, FP, TN, FN be confusion-matrix counts for the positive class PNEUMONIA, π the prevalence of pneumonia, and $s(x) \in [0,1]$ the model probability, thresholded as $\hat{y} = \mathbb{1}[s(x) \geq t]$.

$$\text{Sens} = \text{TPR} = \frac{TP}{TP+FN}, \quad \text{Spec} = 1 - \text{FPR} = \frac{TN}{TN+FP}, \quad \text{BalAcc} = \tfrac{1}{2}(\text{Sens} + \text{Spec}), \quad F_1 = \frac{2TP}{2TP+FP+FN}. \tag{1}$$

$$\text{Acc} = \pi\,\text{Sens} + (1-\pi)\,\text{Spec}, \qquad \text{AUC} = \Pr\big[s(x^+) > s(x^-)\big] + \tfrac{1}{2}\Pr\big[s(x^+) = s(x^-)\big]. \tag{2}$$

Accuracy is a prevalence-weighted average, so it changes with the population screened; sensitivity, specificity and AUC do not.

2 Data audit (Part 1, tasks 1–3)

2.1 Inventory and pixel statistics

The primary data are 5 856 JPEGs (1 583 NORMAL, 4 273 PNEUMONIA). Of these, 283 are stored as RGB but all 5 856 have identical channels, i.e. true greyscale content. The classes differ strongly in image *geometry*, not brightness (Table 1, Table 2, Figure 1).

- Median height is 776 px for PNEUMONIA vs 1 323 px for NORMAL; median aspect ratio 1.49 vs 1.24. The images were evidently assembled from different sources or crops.

Table 1: Inventory of Rohit's original split (`results/audit_summary.md`).

orig. split	NORMAL	PNEUMONIA	total	% pneumonia
test	234	390	624	62.5
train	1341	3875	5216	74.3
val	8	8	16	50.0
all	1583	4273	5856	73.0

Table 2: Can a single metadata number separate the classes? A single-feature AUC far from 0.5 in either direction signals a shortcut. KS = two-sample Kolmogorov–Smirnov statistic, $D = \sup_x |F_{\text{NORMAL}}(x) - F_{\text{PNEUMONIA}}(x)|$.

feature	single-feature AUC	separability $\max(\text{AUC}, 1-\text{AUC})$	KS statistic
mean brightness	0.508	0.508	0.109
contrast (std)	0.302	0.698	0.383
height	0.085	0.915	0.694
width	0.111	0.889	0.638
aspect ratio	0.840	0.840	0.545
area	0.090	0.910	0.685

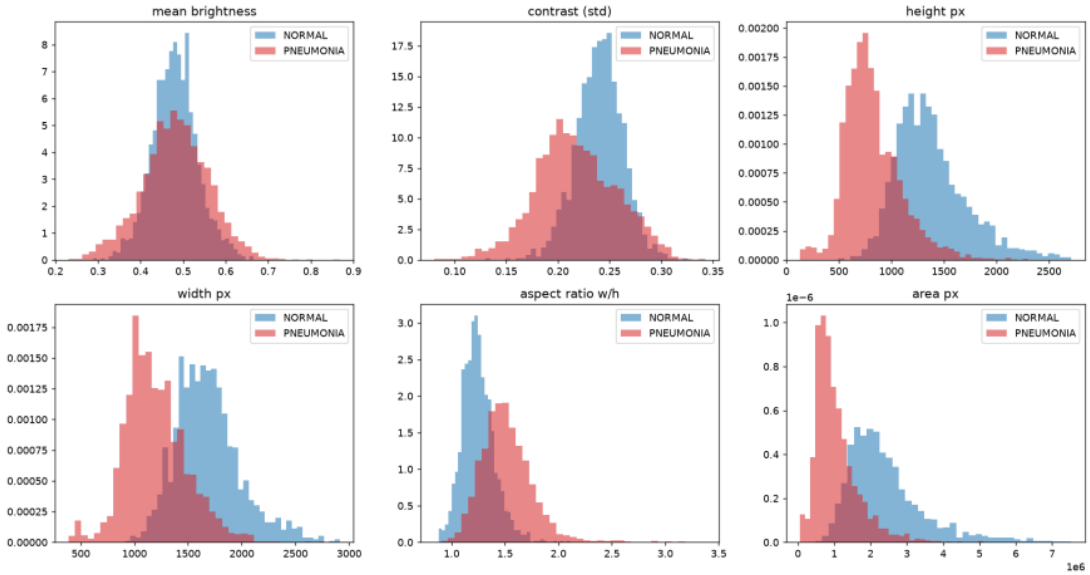


Figure 1: Brightness, contrast and size distributions by class (`figures/audit_class_histograms.png`).

- Height alone gives AUC 0.085 (0.915 with the sign flipped), area 0.090, width 0.111, aspect ratio 0.84. Mean brightness is *not* a shortcut here (0.508) — my initial expectation was wrong; contrast is a weaker one (0.302).
- Implication: a network can score well by recognising crop or scanner format. At the camp, framing and size are set by different hardware, so such cues will not transfer.

2.2 Duplicates and leakage

Exact duplicates. 30 groups of byte-identical files (32 redundant files); none spans two original splits.

Near-duplicates. Each image is reduced to a 64-bit perceptual hash $h \in \{0,1\}^{64}$ (sign of the low-frequency DCT coefficients of a 32×32 thumbnail, relative to their median). Two images are near-duplicates if

$$d_H(h_i, h_j) = \sum_{b=1}^{64} \mathbb{1}\left[h_{i,b} \neq h_{j,b}\right] \leq 4. \tag{3}$$

This flags 239 pairs: 218 within a split and 21 across splits (Table 3); 15 of 624 test images (2.4%) have a train neighbour. The mean pixel correlation of the cross-split pairs is only 0.876, well below what true copies give, so many of these are look-alikes.

Table 3: Near-duplicate pairs by split ($d_H \leq 4$).

pair type	pairs	mean pixel correlation
train vs test	21	0.876
within test	8	0.965
within train	210	0.905

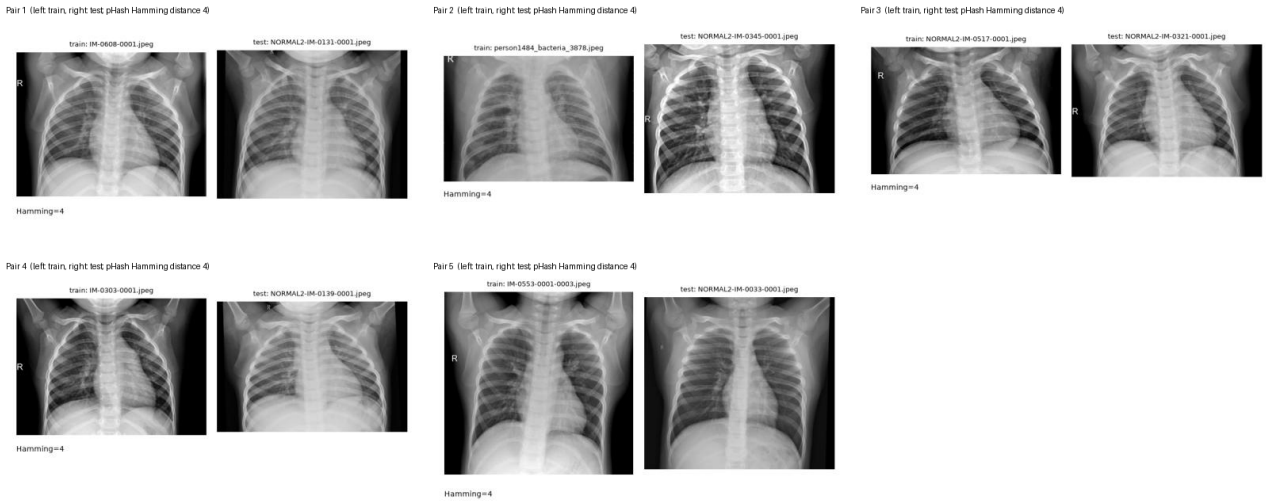


Figure 2: The five closest train/test pairs (left: train, right: test; $d_H = 4$ for all). Pair 2 even crosses the class boundary (a PNEUMONIA training image vs a NORMAL test image).

What the five closest pairs show. Viewed side by side, none of the five is a copy. Pairs 1, 3, 4 and 5 are two NORMAL images (different file-name series) with the same standard frontal framing but different anatomy: the rib spacing, clavicle position and cardiac outline differ, and the "R" side marker sits at a different place in each image. Pair 2 is a PNEUMONIA training image (`person1484_bacteria_3878`) against a NORMAL test image with visibly different brightness and contrast — not the same image, and not the same label. A 4-bit threshold therefore flags look-alike frontal chest X-rays, so *15 of 624 is an upper bound on duplicate leakage*; the firm statement is the exact-duplicate result (none across splits). (This is my visual reading of Figure 2; check it at full resolution.)

Patient overlap. A naive file-name parser reports 264 shared IDs, but these are collisions (e.g. `person100_virus` in train vs `person100_bacteria` in test). With person+pathogen keys for PNEUMONIA and series-aware keys for NORMAL, 0 IDs appear in both train and test. File names cannot prove patient-level separation for NORMAL images, so this is evidence of absence of the *detectable* overlap only.

2.3 The validation set and the replacement design

The original `val/` folder holds 16 images (8 per class). For k successes in n trials the exact (Clopper–Pearson) $(1-\alpha)$ interval is

$$p_L = B^{-1}\left(\tfrac{\alpha}{2}; k, n-k+1\right), \qquad p_U = B^{-1}\left(1-\tfrac{\alpha}{2}; k+1, n-k\right), \tag{4}$$

with B^{-1} the Beta quantile. For $k = n = 16$ this reduces to $p_L = (\alpha/2)^{1/n} = 0.025^{1/16} = 0.794$, so the interval is $[0.794, 1.000]$: a perfect validation score cannot rule out a true accuracy near 79%. Rohit's 98.4% on 624 test images corresponds to about 10 errors (CI $[0.971, 0.992]$ from sampling noise alone).

Replacement. All 5 856 images were pooled and re-split by *group* (patient ID merged with near-duplicate clusters, 2 778 groups, none shared across splits — asserted in `split.py`): a locked test set of 837 images (226 NORMAL / 611 PNEUMONIA, 73.0% pneumonia) and five stratified folds of about 1 004 images (72.9–73.0% pneumonia) on the remaining pool. Fold 0 is the validation set for every decision; the five folds give mean ± std for Part 2; 1 254 images changed split. *Rejected:* a single train/val/test split alone (every decision overfits one slice) and k-fold alone (no untouched test set, hence selection bias).

3 Shortcut hunt and classical baselines (tasks 4–5)

Table 4: Deliberately crippled models, trained on the pool and evaluated on the locked test set (`results/shortcut_summary.md`).

model	acc	bal.acc	recall	spec.	ROC-AUC	PR-AUC
majority class	0.73	0.5	1	0	0.5	0.73
16x16 pixels + LogReg	0.925	0.901	0.953	0.85	0.975	0.991
16x16 pixels + RandForest	0.912	0.871	0.959	0.783	0.968	0.987
lung field masked (border only) + LogReg	0.888	0.844	0.939	0.748	0.923	0.959
metadata only (mean, aspect, area) + LogReg	0.841	0.775	0.918	0.633	0.916	0.97
metadata only (mean, aspect, area) + RandForest	0.858	0.809	0.915	0.704	0.922	0.971

- A 16×16 thumbnail (256 numbers) with logistic regression reaches test AUC 0.975, close to ArogyaNet's 0.989. A thumbnail still shows coarse lung opacity, so this alone is not proof of a shortcut.
- Stronger evidence: with the lung field masked out (border pixels only) AUC is 0.923, and three metadata numbers (mean brightness, aspect ratio, area) alone give 0.916–0.922. *Information outside the lungs identifies the class.*
- Consequence: high accuracy on this dataset does not show that a model reads lung tissue.



Figure 3: Examples of the lung-masked inputs used for the border-only model.

Table 5: Baselines (trained on folds 1–4, validated on fold 0, then locked test) — `results/baselines_summary.md`.

model	split	acc	bal.acc	precision	recall	spec.	f1	ROC-AUC	PR-AUC
majority class	val	0.729	0.5	0.729	1	0	0.843	0.5	0.729
majority class	test	0.73	0.5	0.73	1	0	0.844	0.5	0.73
HOG + LogReg	val	0.943	0.929	0.962	0.96	0.897	0.961	0.988	0.995
HOG + LogReg	test	0.939	0.932	0.968	0.948	0.916	0.958	0.981	0.993
LBP + LogReg	val	0.934	0.909	0.946	0.964	0.853	0.955	0.977	0.991
LBP + LogReg	test	0.912	0.889	0.941	0.938	0.841	0.939	0.959	0.984

The bar for deep learning is high: HOG + logistic regression already reaches test AUC 0.981 and balanced accuracy 0.932, so the best deep model improves AUC by less than two points.

4 Models and ablations (tasks 6–9)

Table 6: Mean ± std over 3 seeds on the locked test set, threshold 0.5. Decisions were made on validation fold 0; test numbers are for reporting (`results/summary_runs.md`).

tag	seeds	params	ROC-AUC	PR-AUC	bal.acc	sens.	spec.
resnet18	3	11177025	0.995 ± 0.000	0.998 ± 0.000	0.968 ± 0.007	0.981 ± 0.001	0.954 ± 0.016
mobilenetv3	3	927585	0.994 ± 0.001	0.998 ± 0.000	0.962 ± 0.007	0.981 ± 0.011	0.944 ± 0.024
arogyanet	3	242319	0.988 ± 0.001	0.996 ± 0.000	0.945 ± 0.006	0.954 ± 0.012	0.937 ± 0.003
arogyanet_nores	3	206159	0.985 ± 0.001	0.994 ± 0.000	0.932 ± 0.003	0.952 ± 0.003	0.913 ± 0.007
arogyanet_nose	3	217827	0.988 ± 0.001	0.996 ± 0.000	0.944 ± 0.006	0.954 ± 0.008	0.934 ± 0.004
arogyanet_meta	3	242677	0.990 ± 0.001	0.996 ± 0.001	0.949 ± 0.007	0.965 ± 0.004	0.934 ± 0.018
arogyanet_res96	3	242319	0.985 ± 0.003	0.994 ± 0.001	0.934 ± 0.003	0.959 ± 0.006	0.909 ± 0.007
arogyanet_aug_none	3	242319	0.982 ± 0.001	0.993 ± 0.001	0.934 ± 0.006	0.969 ± 0.003	0.900 ± 0.009
arogyanet_aug_naive	3	242319	0.988 ± 0.002	0.996 ± 0.001	0.946 ± 0.004	0.959 ± 0.007	0.934 ± 0.012
arogyanet_tuned	3	242319	0.989 ± 0.001	0.996 ± 0.000	0.953 ± 0.003	0.956 ± 0.005	0.950 ± 0.007
arogyanet_tuned_cw	3	242319	0.984 ± 0.004	0.994 ± 0.001	0.933 ± 0.014	0.939 ± 0.005	0.928 ± 0.026
arogyanet_tuned_clahe	3	242319	0.989 ± 0.001	0.996 ± 0.001	0.950 ± 0.003	0.968 ± 0.006	0.932 ± 0.011

4.1 Transfer baseline

Expectation, written before running (`reports/hypotheses.md`, commit `4f614ff`, 2026-11-08 15:01 +0530): ImageNet low-level filters (edges, textures) transfer to radiographs, high-level object features do not, so pretrained backbones should be at least as good as ArogyaNet without a wide margin. *Result:* ResNet18 test AUC 0.995 and MobileNetV3 0.994 vs ArogyaNet 0.988; balanced accuracy 0.968 / 0.962 vs 0.945. The expectation held: the gap is small in AUC but larger in balanced accuracy. ResNet18 has 11.2 M parameters (about 45 MB in float32, over the 25 MB channel); MobileNetV3-small has 0.93 M (about 4 MB).

4.2 ArogyaNet

242 319 parameters (limit 2 M), no pretrained weights. A stem convolution, five blocks, global average pooling and a linear head. Each block stacks two depthwise-separable convolutions, a squeeze-and-excitation (SE) unit (decision D2) and a residual shortcut (decision D1):

$$y = \mathrm{ReLU6}\big(\underbrace{\mathrm{SE}\big(F(x)\big)}_{\text{D2}} + \underbrace{S(x)}_{\text{D1}}\big), \qquad \mathrm{SE}(u)_c = \sigma\big(W_2\,\mathrm{ReLU}(W_1 z)\big)_c\, u_c, \quad z_c = \frac{1}{HW}\sum_{h,w} u_{c,h,w}. \tag{5}$$

Depthwise-separable convolutions cut the parameters of a $k \times k$ convolution from $k^2 C_{\text{in}} C_{\text{out}}$ to $k^2 C_{\text{in}} + C_{\text{in}} C_{\text{out}}$ (about $8\times$ fewer for $k = 3$, $C = 128$), which keeps the CPU latency low. The architecture table is in `results/arch_table.md`.

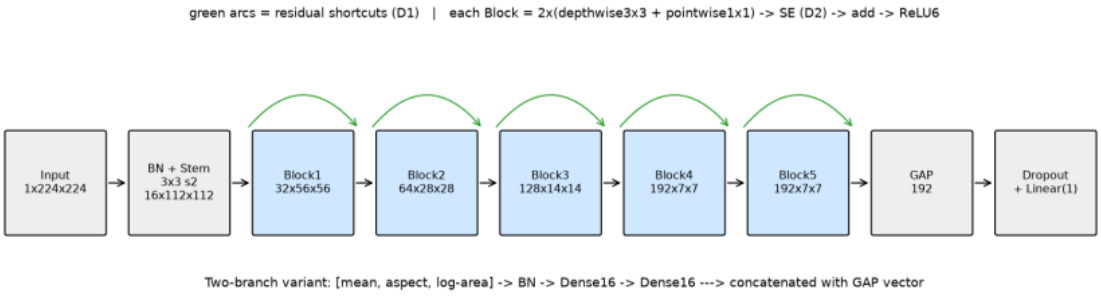


Figure 4: ArogyaNet. Green arcs: residual shortcuts (D1). Each block: 2×(depthwise 3×3 + pointwise 1×1) → SE (D2) → add → ReLU6.

Two-branch experiment. Adding mean intensity, aspect ratio and area through a small dense branch (concatenated before the classifier) did not improve results beyond noise (test AUC 0.990 vs 0.988). This does not show the metadata is useless: alone it separates the classes with AUC about 0.92 (Table 4), so the image branch can already read equivalent cues

Table 7: Hypotheses written before the results vs what was observed. Noise rule used throughout: a difference Δ is "inside noise" if $|\Delta| < \sqrt{s_a^2 + s_b^2}$ (standard deviations over 3 seeds); with only 3 seeds this rule is crude.

Hypothesis	Observed	Verdict
D1 residual connections: small gain, probably inside noise	Removing them: AUC -0.004 (spread ± 0.001), balanced accuracy -0.013 (± 0.007); on validation mostly inside noise	Small but consistent on test; weak on validation
D2 squeeze-excitation: about $+1$ pp PR-AUC / specificity	Removing it: every metric inside noise (AUC 0.988 vs 0.988); costs 24k parameters	**Not supported**
Pretrained $\geq$ ArogyaNet, not by a wide margin	ResNet18 / MobileNetV3 AUC $+0.007$ / $+0.006$, balanced accuracy $+0.022$ / $+0.017$	Supported
Metadata branch helps internally	AUC $+0.002$ (± 0.002); all metrics inside noise	**Not supported**
Flip augmentation changes nothing internally	All metrics inside noise	Supported

Table 8: Ablations on the locked test set: change relative to the reference (Δ = variant $-$ reference, $\pm$ combined spread). All values, including validation: Table 18.

ablation	ΔAUC	verdict	Δbal.acc	verdict
D1: remove residual connections	0.985 ± -0.004	±0.001	0.932 ± -0.013	±0.007
D2: remove squeeze-excitation	0.988 ± 0	±0.001	0.944 ± -0.001	±0.009
add metadata branch (two-branch)	0.99 ± 0.002	±0.002	0.949 ± 0.004	±0.009
resolution 224 → 96 px	0.985 ± -0.003	±0.003	0.934 ± -0.012	±0.006
augmentation: safe → none	0.982 ± -0.006	±0.001	0.934 ± -0.011	±0.008
augmentation: safe → naive (+ h-flip)	0.988 ± 0	±0.002	0.946 ± 0.001	±0.007
ResNet18 (pretrained) vs ArogyaNet	0.988 ± -0.007	±0.001	0.945 ± -0.022	±0.009
class weights (tuned model)	0.984 ± -0.005	±0.004	0.933 ± -0.02	±0.014
CLAHE preprocessing (tuned model)	0.989 ± 0	±0.001	0.95 ± -0.003	±0.004

(framing, borders, intensity layout). We therefore cannot claim the model learned something specific about these patients — only that its signal is not distinguishable from such cues. At the camp they would not hold: the external hospital's images are all 299×299 (KS statistic for width $= 1.0$), and framing and size depend on the scanner.

Resolution and augmentation.

- 96 px vs 224 px: inside noise on validation; on test AUC -0.003 (± 0.003) and balanced accuracy -0.012 (± 0.006). A cheaper 96 px model loses little.
- No augmentation vs small rotations/shifts/zoom: no augmentation is worse on test (AUC -0.006 ± 0.001, PR-AUC -0.003, balanced accuracy -0.011).
- **The medically wrong augmentation is the horizontal flip** (and the vertical flip). It places the heart on the right — dextrocardia, a rare and, for this population, implausible anatomy — and swaps the meaning of the left/right laterality marker. Strictly, the pneumonia label is unchanged by a flip; what is corrupted is the anatomical context the image was labelled in. In our data the flip did not hurt measurably (inside noise on every metric): the benchmark metric is blind to laterality, so this is an argument from anatomy, not an experimental finding.

5 Imbalance, threshold and cost (Part 2.1)

Class weights vs threshold moving. With N_+ and N_- training positives and negatives, the class-weighted loss multiplies the positive term by $w = N_-/N_+$:

$$\mathscr{L} = -\frac{1}{N}\sum_i \Big[w\, y_i \log p_i + (1-y_i)\log(1-p_i) \Big], \qquad p_i = \sigma(z_i). \tag{6}$$

This changes the *network*: relative weight distance 0.25–0.28 from the unweighted model, validation probability rank correlation 0.91–0.97, test AUC 0.984 vs 0.989. Moving the threshold t leaves the network, its probabilities and its AUC unchanged and only changes the confusion matrix (Table 19, Appendix). At a validation-chosen "95% sensitivity" threshold, test sensitivity was only 0.91–0.94: *a sensitivity target set on validation does not transfer exactly*.

Cost-optimal threshold. A missed pneumonia costs $c_{FN} = 20$, a false alarm $c_{FP} = 1$; camp prevalence is $\pi = 0.10$.

Because TPR and FPR do not depend on prevalence, the camp's expected cost per child can be evaluated on a dataset with a different class mix:

$$\mathscr{C}(t) = \pi\, c_{FN}\,\big(1-\mathrm{TPR}(t)\big) + (1-\pi)\, c_{FP}\,\mathrm{FPR}(t), \qquad \hat{t} = \arg\min_t \mathscr{C}(t). \tag{7}$$

Predictive values at the camp follow from Bayes' rule:

$$\mathrm{PPV} = \frac{\pi\,\mathrm{TPR}}{\pi\,\mathrm{TPR} + (1-\pi)\,\mathrm{FPR}}, \qquad \mathrm{NPV} = \frac{(1-\pi)(1-\mathrm{FPR})}{(1-\pi)(1-\mathrm{FPR}) + \pi\,(1-\mathrm{TPR})}. \tag{8}$$

At the chosen point (TPR $= 0.949$, FPR $= 0.044$): PPV $= 0.0949/(0.0949+0.0398) = 0.704$ and $\mathscr{C} \approx 0.10 \cdot 20 \cdot 0.051 + 0.90 \cdot 0.044 \approx 0.14$ per child, i.e. about 141 per 1 000 children, against $\pi c_{FN} = 2.0$ per child for "always NORMAL" (2 000 per 1 000) and $(1-\pi)c_{FP} = 0.9$ for "refer everyone".

The threshold chosen on validation is 0.515, close to 0.5. This is expected. For a calibrated posterior the camp-optimal threshold is $t^* = c_{FP}/(c_{FP}+c_{FN}) = 1/21 \approx 0.048$ at camp prevalence; the model was trained at prevalence $\pi_t \approx 0.73$, so on its own probability scale the threshold is shifted by the prior odds ratio:

$$\frac{t}{1-t} = \frac{t^*}{1-t^*} \cdot \frac{\pi_t/(1-\pi_t)}{\pi_c/(1-\pi_c)} \;\Rightarrow\; \frac{t}{1-t} = \frac{1}{20} \cdot \frac{0.73/0.27}{0.10/0.90} \approx 1.22 \;\Rightarrow\; t \approx 0.55. \tag{9}$$

The 20× miss cost pushes the threshold down, while moving from 73% to 10% prevalence pushes it up; the two nearly cancel, and the analytic value (0.548) agrees with the empirical one. The model's probabilities are very confident, so the cost curve is flat around the optimum (Figure 5).

Table 9: Test-set confusion and camp-prevalence metrics at three thresholds (`results/threshold_cost.md`); PPV, NPV and cost per child are re-weighted to 10% prevalence.

rule	thr	tn	fp	fn	tp	sens	spec	ppv	npv	cost/child
default 0.5	0.5	216	10	30	581	0.951	0.956	0.7048	0.9943	0.138
cost-optimal (chosen on val)	0.5152	216	10	31	580	0.949	0.956	0.7045	0.9941	0.1413
prior-shift Bayes (assumes calibration)	0.5477	216	10	32	579	0.948	0.956	0.7041	0.9939	0.1446

Table 10: Per 1 000 children at the camp (100 sick, 900 healthy); cost = 20× missed +1× false alarm (`results/ceo_arithmetic.md`). Accuracy cannot separate the third and fourth rows, whose costs differ 20-fold.

scenario	missed	false alarms	accuracy	sensitivity	PPV	NPV	cost
Always say NORMAL (does nothing)	100	0	90.0%	0.0%	n/a	90.0%	2000
Pilot model: ~70% of healthy flagged	0	630	37.0%	100.0%	13.7%	100.0%	630
Hypothetical '99% accuracy', all 10 errors = missed cases	10	0	99.0%	90.0%	100.0%	98.9%	200
Hypothetical '99% accuracy', all 10 errors = false alarms	0	10	99.0%	100.0%	90.9%	100.0%	10
ArogyaNet @ cost-optimal threshold 0.515 (locked test)	5.1	39.8	95.5%	94.9%	70.4%	99.4%	141.3

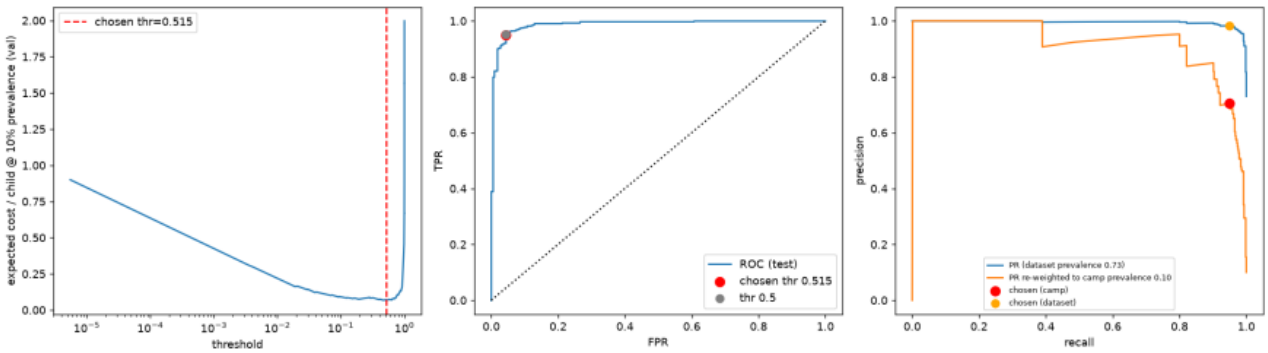


Figure 5: Left: expected cost per child vs threshold (validation). Middle: ROC (test) with the chosen point and $t = 0.5$. Right: precision–recall (test) and the same curve re-weighted to camp prevalence.

6 Evaluation and cross-validation (Part 2.2)

At the chosen threshold on the locked test set the confusion matrix is $TN = 216$, $FP = 10$, $FN = 31$, $TP = 580$. For $K = 5$ folds each metric is reported as $\bar{m} = \frac{1}{K}\sum_k m_k$ and $s = \sqrt{\frac{1}{K-1}\sum_k (m_k - \bar{m})^2}$ (Table 11).

Table 11: 5-fold stratified, group-aware CV at the fixed threshold 0.515 vs the single locked-test model (`results/cv_table.md`).

	acc	precision	recall	specificity	f1	ROC-AUC	PR-AUC	bal.acc
5-fold CV	0.966 ± 0.004	0.977 ± 0.005	0.977 ± 0.004	0.938 ± 0.014	0.977 ± 0.003	0.992 ± 0.001	0.996 ± 0.002	0.957 ± 0.007
locked test (single model)	0.951	0.983	0.949	0.956	0.966	0.989	0.995	0.953

Cross-validated sensitivity (0.977 ± 0.004) is higher than the locked-test value (0.949). The three seeds' test sensitivities range from 0.941 to 0.964 and the reported seed was selected on validation, so part of the gap is seed variation; the threshold was also chosen on fold 0, a small optimistic bias for CV.

Why accuracy alone is the wrong headline. By Eq. (2), $\text{Acc} = \pi\,\text{Sens} + (1-\pi)\,\text{Spec}$: at $\pi = 0.1$ a model that never flags anyone has Sens $= 0$, Spec $= 1$ and Acc $= 0.9$ while missing every sick child; and accuracy weighs a miss like a false alarm although a miss is about $20\times$ costlier.

7 Tuning and regularisation (Part 2.3)

Budget: random search, 12 trials $\times$ 8 epochs, one seed, validation fold 0. Space: learning rate log-uniform in $[10^{-4}, 3\times 10^{-3}]$, batch size $\{16, 32, 64\}$, dropout $\{0, 0.1, 0.3, 0.5\}$, weight decay log-uniform in $[10^{-6}, 10^{-2}]$; selection by lowest validation loss. Chosen: learning rate 8.4×10^{-4}, batch 32, dropout 0.1, weight decay about 10^{-6}. The best trials differ very little (Table 12), the chosen weight decay is almost zero, and the tuned model is not better than the default beyond noise (test AUC 0.989 vs 0.988): the search is best read as confirming the defaults.

Table 12: Top five search trials (`results/tuning.md`).

trial	lr	bs	dropout	wd	best_val_loss	val_roc_auc	val_pr_auc	best_epoch
5	0.00084	32	0.1	0	0.09688	0.99107	0.99595	6
0	0.00036	16	0.3	0.00085	0.09999	0.99036	0.99331	8
3	0.00095	16	0.1	0.00077	0.10044	0.99058	0.99109	8
4	0.00256	32	0.5	0.00931	0.10169	0.9903	0.99094	7
8	0.00085	64	0.5	0.00023	0.1018	0.98991	0.9951	7

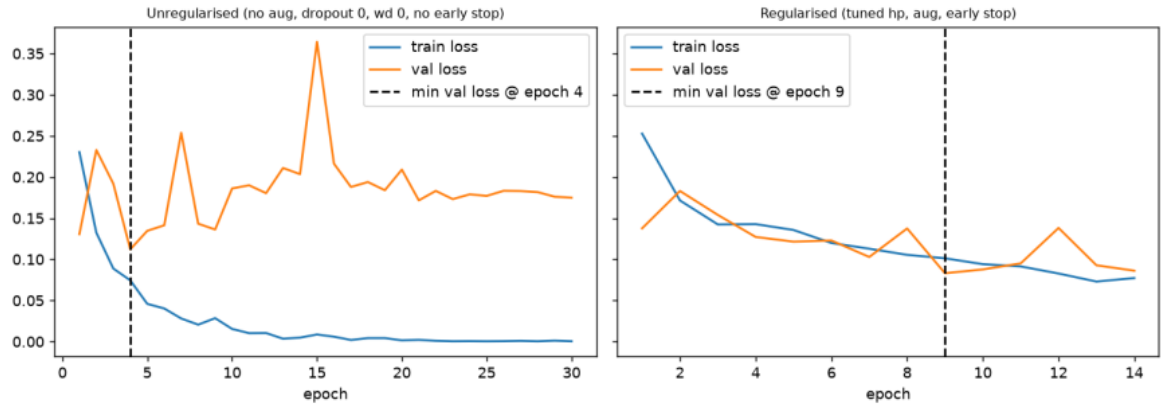


Figure 6: Learning curves. Left: unregularised (no augmentation, dropout 0, weight decay 0, no early stopping). Right: regularised (tuned hyper-parameters, augmentation, early stopping). Dashed line: epoch of minimum validation loss.

Overfitting begins right after epoch 4 in the unregularised run (validation loss 0.112, rising to 0.175 by epoch 30 while training loss falls to 0.0004). The regularised run reached its best epoch at 9 (validation loss 0.083) and early stopping ended it at epoch 14 of 30, saving 16 epochs and avoiding a validation-loss rise of 0.063. The two runs differ in several settings at once, so the saving cannot be credited to early stopping alone.

8 Ensembles (Part 2.4)

A soft-voting ensemble of M members averages their probabilities, $\bar{s}(x) = \frac{1}{M}\sum_{m=1}^{M} s_m(x)$; it helps when members make *uncorrelated* errors.

Table 13: Single members, soft-voting ensembles and tree models on penultimate-layer CNN features (full list: `results/ensemble_summary.md`). The "all members" row is mathematically identical to the family average (3 seeds per family) and is omitted.

model	val ROC	test ROC	test PR	test bal.acc	test sens.	test spec.
arogyanet_s0	0.9948	0.9889	0.9958	0.9396	0.9411	0.9381
resnet18_s0	0.9986	0.9952	0.9982	0.9747	0.9804	0.969
mobilenetv3_s0	0.9967	0.9934	0.9975	0.9547	0.9935	0.9159
SOFT-VOTE arogyanet seeds	0.9944	0.99	0.9962	0.9486	0.9591	0.9381
SOFT-VOTE family-average (arogyanet+resnet18+mobilenetv3)	0.9982	0.9955	0.9983	0.9703	0.9804	0.9602
RandomForest on CNN features	0.9914	0.9801	0.9877	0.9504	0.9673	0.9336
AdaBoost on CNN features	0.9894	0.9814	0.9909	0.9474	0.9656	0.9292

The three-family soft vote (test AUC 0.9955) equals its best member (ResNet18, 0.9952) within seed noise: member probabilities correlate at 0.96 and member errors at 0.52, so averaging removes little variance. Random forest and AdaBoost on the penultimate-layer features were worse than the network's own head (AUC 0.980 and 0.981 vs 0.989), consistent with fitting on embeddings the CNN has already fitted.

9 Feature-space analysis (Part 2.5)

Penultimate embeddings $z \in \mathbb{R}^{192}$ of the test set were reduced with PCA and t-SNE (Figure 7). k-means with $k = 2$ was scored against the true labels with purity and the adjusted Rand index,

$$\text{purity} = \frac{1}{N}\sum_{c}\max_{j}|C_c \cap L_j|, \qquad \text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]}, \tag{10}$$

obtaining ARI 0.828 and purity 0.957 (majority-class purity 0.730): the embedding separates the classes without using labels, although the network was trained on them.

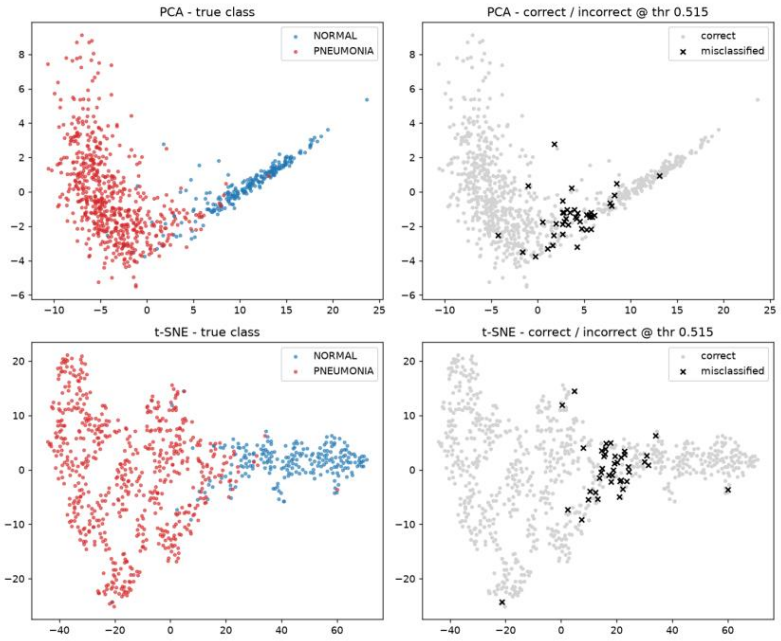


Figure 7: PCA (top) and t-SNE (bottom) of test embeddings, coloured by true class (left) and by correct/incorrect at $t = 0.515$ (right).

What the mistakes have in common.

Table 14: Errors vs correct predictions on the locked test set (model `arogyanet_tuned`, seed 0). `pct_bacterial` / `pct_virus`: share of the group that is bacterial / viral pneumonia.

kind	n	mean_brightness	contrast_std	mean_area	mean_aspect	mean_prob	pct_bacterial	pct_virus
correct	796	0.48	0.224	1.37849e+06	1.439	0.716	49.874	22.99
false alarm (FP)	10	0.521	0.235	1.76767e+06	1.276	0.843	0	0
missed pneumonia (FN)	31	0.473	0.24	1.59949e+06	1.398	0.199	38.71	61.29



Figure 8: Misclassified test images: three missed pneumonias (FN, $p \leq 0.01$) and three false alarms (FP, $p = 0.96$–1.00) (`figures/misclassified_examples.png`).

- *Missed pneumonia (31).* 61% are viral and 39% bacterial, about twice the viral share among all pneumonia cases (roughly 31%). They are *confidently* wrong (mean probability 0.20; the three shown have $p \leq 0.01$), so lowering the threshold would not recover most of them. At thumbnail resolution the three shown have well-aerated, fairly clear-looking lung fields with no obvious dense opacity (I am not a clinician; this is a visual impression only), which fits viral pneumonia producing subtle changes. One of them is also a differently framed, washed-out image.
- *False alarms (10).* All are NORMAL images (mean probability 0.84), slightly brighter and larger than average (Table 14). The three shown include a tightly cropped, magnified chest that fills the frame, and two with prominent overlays (a large "R" marker and a timestamp-like text block): non-lung content the model may be reacting to. With $n = 10$ this is suggestive only.
- *In embedding space (Figure 7).* The misclassified points concentrate where the NORMAL and PNEUMONIA regions overlap in the 2-D projection, rather than being scattered over either cluster.

Descriptions are my reading of Figure 8; please check them against the full-resolution images in `figures/`.

10 External-hospital generalisation, run once (Part 2.6)

The external folder holds 11 537 images (10 192 Normal, 1 345 Viral Pneumonia; 11.7% pneumonia), all used. It is touched only by `external_eval.py`, which refuses a second run; thresholds were fixed on internal validation data. For two samples with empirical distributions F_1, F_2 the Kolmogorov–Smirnov statistic is $D = \sup_x |F_1(x) - F_2(x)|$ (0 = identical, 1 = disjoint).

Table 15: Internal test vs external hospital (`results/external_summary.md`); camp-PPV / NPV / cost are re-weighted to 10% prevalence.

model	data	thr	roc_auc	pr_auc	sens	spec	bal_acc	acc	camp_ppv	camp_npv	camp cost
final (ArogyaNet)	internal test	0.515	0.989	0.995	0.949	0.956	0.953	0.951	0.704	0.994	0.141
final (ArogyaNet)	EXTERNAL hospital	0.515	0.899	0.423	0.953	0.726	0.84	0.753	0.279	0.993	0.34
CLAHE fix (retrained)	internal test	0.707	0.989	0.996	0.966	0.925	0.945	0.955	0.588	0.996	0.136
CLAHE fix (retrained)	EXTERNAL hospital	0.707	0.852	0.289	0.97	0.668	0.819	0.703	0.245	0.995	0.36

- **Drop:** ROC-AUC 0.989 → 0.899 (−0.090), specificity 0.956 → 0.726, balanced accuracy 0.953 → 0.840. Sensitivity stayed 0.953, but the external positives are viral pneumonia in adults, so this does not show that sensitivity generalises.
- **At camp prevalence** PPV fell from 0.70 to 0.28 and the expected cost per child rose from 0.14 to 0.34 — still better than flagging nobody (2.0) or everybody (0.9). PR-AUC is 0.423 against a chance level of about 0.12 (the external prevalence).
- **Distribution mismatch** (pixel statistics from Part 1): every external image is 299×299, whereas internal images average about $1\,400 \times 1\,700$ (NORMAL) and $820 \times 1\,200$ (PNEUMONIA) pixels (KS: height 0.987, width 1.0);

the contrast distribution differs moderately (KS 0.238) and mean brightness little (0.122) (Figure 9). Internally, geometry alone separated the classes (Table 2), a cue that cannot exist in a uniformly sized external set. Other differences are population (children 1–5 years vs adults), projection (AP vs PA) and scanner or processing. These are consistent explanations; we did not test them causally.

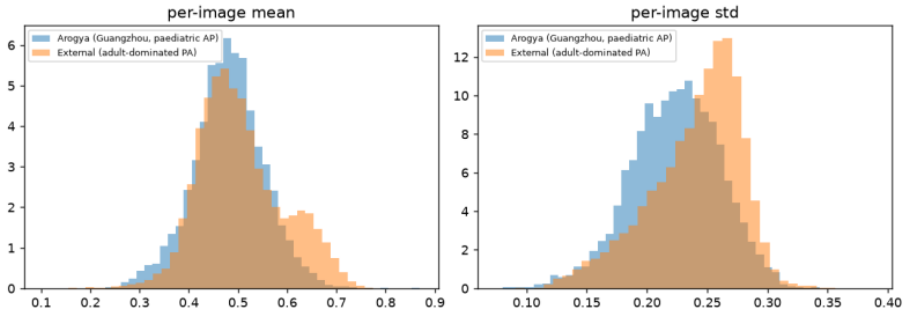


Figure 9: Per-image mean intensity and contrast: internal data vs external hospital.

Fix attempted: CLAHE preprocessing, retrained on internal data. It did *not* help: external AUC 0.852 (vs 0.899) and specificity 0.668 (vs 0.726); internally it was inside noise. Not adopted.

11 Shipping (Part 2.7)

Decision band. From validation data only (`set_band.py`):

$$t_{\text{low}} = \max\{t : \text{TPR}_{\text{val}}(t) \geq 0.98\} = 0.461, \qquad t_{\text{high}} = \min\{t : \text{FPR}_{\text{val}}(t) \leq 0.01\} = 0.885. \tag{11}$$

$s < t_{\text{low}} \Rightarrow$ NORMAL; $t_{\text{low}} \leq s < t_{\text{high}} \Rightarrow$ LOW CONFIDENCE, refer to a clinician; $s \geq t_{\text{high}} \Rightarrow$ PNEUMONIA suspected. Only the NORMAL zone clears a child (Table 16).

Table 16: Decision zones on the locked test set, per 1 000 camp children (`results/band.md`).

zone	test images	per 1,000 children	of which sick	of which healthy
NORMAL (cleared)	245	861	4.9	856.2
LOW CONFIDENCE (refer)	43	34	5.9	27.9
PNEUMONIA suspected (refer)	549	105	89.2	15.9

Validation had suggested 98% sensitivity for the NORMAL cut; on the test set the referred group has sensitivity 0.951 (95% CI 0.931–0.967). At camp prevalence the NORMAL zone has NPV 0.994, 13.9% of children are referred, and the PNEUMONIA zone has PPV 0.848.

Operating constraints. End-to-end single-image latency on this laptop's CPU (file read, resize, forward pass, out-of-distribution check): median 128.0 ms, 95th percentile 138.4 ms (limit 200 ms) — pass, but with modest margin on a slower camp laptop. Model file 1.05 MB (limit 25 MB) — pass.

Non-X-ray inputs. Three guards run before any answer is shown: a blank-image check, a colour check, and an embedding-distance check. With training-set embedding mean μ and regularised covariance Σ the Mahalanobis distance is

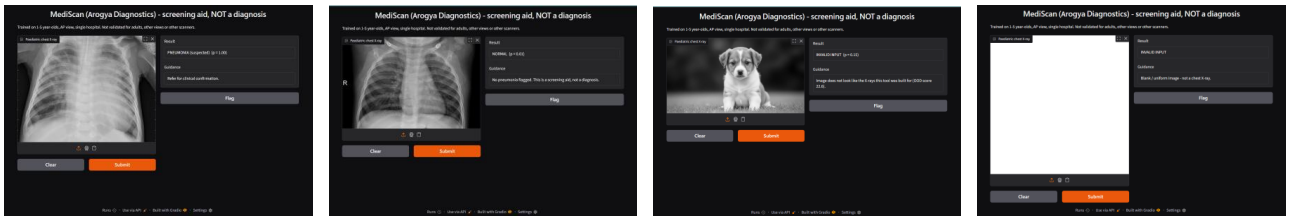
$$d_M(z) = \sqrt{(z-\mu)^\top (\Sigma + \lambda I)^{-1} (z-\mu)}, \qquad \text{reject if } d_M(z) > \tau. \tag{12}$$

A first version ($\tau = 25.0$, from 1.3× the validation 99th percentile) let a greyscale puppy photo through as NORMAL ($d_M = 22.7$; real X-rays reached 24.0). Lowering τ to 20.0, chosen from the validation distances (real-X-ray 99th percentile 17.4–19.3), rejects it; the price is that roughly 1% of real X-rays may be rejected and need to be retaken. This is a mitigation, not a guarantee. (The auto-generated `results/deploy.md` still prints the old 25.0; `models/final_meta.json` holds 20.0.)

Must not be used for: diagnosis; adults; PA or non-AP views; other scanners or hospitals without re-validation; children with symptoms who should be examined regardless; any setting without a clinician in the loop.

Table 17: Rejection test (`results/rejection_test.md`): no non-X-ray received a medical answer; the six real X-ray controls did.

input	outcome	guard / probability
blank_black, blank_white, uniform_grey	INVALID INPUT	blank-image guard (std < 0.03)
colour_blobs, dog.jpg (puppy photo)	INVALID INPUT	colour guard
dog_grey.jpg (greyscale puppy)	INVALID INPUT	$d_M = 22.7 > 20$
gaussian_noise_grey, grey_stripes	INVALID INPUT	$d_M = 115.1, 60.9$
6 real X-rays (control)	answered	PNEUMONIA 0.996, 1.000, 1.000, 1.000; NORMAL 0.000, 0.002



(a) Pneumonia X-ray **(b)** Normal X-ray **(c)** Greyscale puppy **(d)** Blank image

Figure 10: Gradio demo (`app.py`): real X-rays get a class and probability; the puppy and the blank image are rejected as INVALID INPUT.

12 Limitations and honest notes

- *Selection:* hyper-parameters, the threshold and the band were chosen on validation fold 0 only; the test set was not used for any decision but was evaluated for every run.
- The deployed model is ArogyaNet because the external test was run on it. MobileNetV3 is stronger in-distribution (AUC 0.994) and small enough to ship, but its external performance is unknown and the external set cannot be reused.
- Three seeds give crude noise estimates; the "inside noise" rule is a heuristic, not a significance test.
- File-name IDs are weak evidence about patients; the zero-overlap finding is not proof of patient-level separation. The perceptual-hash threshold over-counts look-alikes, so 2.4% is an upper bound on duplicate leakage.
- The 98.4% of the previous model cannot be reproduced; we do not claim to know how it was obtained.
- My descriptions of the images in Figures 2 and 8 are visual impressions, not clinical readings.

13 Reproducibility

`pip install -r requirements.txt`; `python run_all.py` runs every script in order (audit, split, shortcut, baselines, arch_report, training, aggregate, tune, threshold_cost, ceo_numbers, cv_eval, ensemble, embeddings, make_final, reject_test, external_eval, build_report). `set_band.py` and the report builder are run afterwards; the out-of-distribution threshold in `models/final_meta.json` was set to 20.0 after `ood_check.py`.

A All ablations (validation and test)

Table 18: Change relative to the reference (Δ = variant − reference, $\pm$ combined spread) and verdict (noise = inside noise). Source: `results/summary_runs.md`.

ablation	val ΔAUC		test ΔAUC		val Δbal.acc		test Δbal.acc	
D1: remove residual connections	0.991 ± -0.002	±0.002	0.985 ± -0.004	±0.001	0.947 ± -0.019	±0.016	0.932 ± -0.013	±0.007
D2: remove squeeze-excitation	0.992 ± -0.001	±0.002	0.988 ± 0	±0.001	0.959 ± -0.007	±0.006	0.944 ± -0.001	±0.009
add metadata branch (two-branch)	0.994 ± 0.001	±0.002	0.99 ± 0.002	±0.002	0.964 ± -0.002	±0.007	0.949 ± 0.004	±0.009
resolution 224 → 96 px	0.992 ± -0.001	±0.002	0.985 ± -0.003	±0.003	0.961 ± -0.005	±0.007	0.934 ± -0.012	±0.006
augmentation: safe → none	0.991 ± -0.002	±0.002	0.982 ± -0.006	±0.001	0.952 ± -0.014	±0.007	0.934 ± -0.011	±0.008
augmentation: safe → naive (+ h-flip)	0.992 ± -0.001	±0.002	0.988 ± 0	±0.002	0.958 ± -0.008	±0.008	0.946 ± 0.001	±0.007
ResNet18 (pretrained) vs ArogyaNet	0.993 ± -0.005	±0.002	0.988 ± -0.007	±0.001	0.966 ± -0.011	±0.007	0.945 ± -0.022	±0.009
class weights (tuned model)	0.99 ± -0.003	±0.003	0.984 ± -0.005	±0.004	0.955 ± -0.014	±0.014	0.933 ± -0.02	±0.014
CLAHE preprocessing (tuned model)	0.994 ± 0.001	±0.001	0.989 ± 0	±0.001	0.963 ± -0.006	±0.006	0.95 ± -0.003	±0.004

B Class weights vs threshold moving, per seed

Table 19: Per-seed comparison of the class-weighted and the unweighted model (`results/threshold_cost.md`, section A).

	0	**1**	**2**
seed	0	1	2
weight_rel_L2_distance	0.2511	0.2453	0.2849
val_AUC_plain	0.9939	0.9924	0.9925
val_AUC_weighted	0.9872	0.9921	0.9919
mean_prob_plain	0.716	0.7185	0.7241
mean_prob_weighted	0.7105	0.7049	0.7132
spearman_val_probs	0.9114	0.9676	0.9504
test_spec_at_95sens_threshold_moved	0.9735	0.969	0.9425
test_sens_at_95	0.9067	0.928	0.9394
test_spec_at_95sens_weighted	0.9204	0.9558	0.9646
test_sens_w	0.9231	0.9231	0.9133